Abdulkader Alturkmani

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Education

University of Toronto Sep 2023 - Nov 2027

Bachelor of Science, Geospatial Data Science Major, Applied Statistics & Computer Science Minor

- **Relevant Coursework**: Spatial Data Science, Introduction to Computer Science, Probability and Statistics, Linear Algebra, Introduction to Geospatial Data Science, Mathematical Proofs.

Projects

Viewshed & Hiking Route Analysis | *ArcGIS Pro, ModelBuilder*

- Compared three viewpoints in Pipestone Valley using geodesic viewshed analysis, identifying a site with 153.14 km^2 of visible terrain.
- Built a ModelBuilder workflow weighting slope and landcover costs equally to generate a least-cost hiking route around a trail closure.
- Applied Tobler's Hiking Function to estimate walking speeds and created maps comparing terrain costs, visibility, and the proposed route.

Treemap Filesystem Visualizer | *Python, Data Structures*

- Built an interactive Python treemap that represents files and folders as proportional rectangles, revealing storage distribution across directory trees.
- Represented nested directories with tree data structures to organize file sizes and preserve parent-child relationships.
- Applied recursive rectangle subdivision to allocate screen space by subtree size, making large files and folders easy to compare.

Java Paint Application with AI Integration | *Java, JavaFX, Ollama API*

- Developed a JavaFX drawing application for creating and editing geometric shapes with customizable colors and stroke widths.
- Implemented undo/redo and clipboard operations so users could revise drawings, reuse shapes, and recover earlier editing states.
- Integrated the Ollama API to support prompt-based image generation and editing within the drawing interface.

Experience

Frontend Web Design Intern @ Pepperdean | *WordPress, Elementor, Figma* Mar 2026 - May 2026

- Analyzed user engagement and on-page behavior to guide a full header redesign, layout restructuring, and visual design drafts.
- Ran keyword and competitor analysis to shape SEO and AEO strategy, translating findings into concrete changes on the modernization roadmap.
- Collaborated async via Slack through iterative, feedback-driven revision cycles, tracking design iterations against project goals.

Skills

Geospatial Analysis: ArcGIS Pro, ModelBuilder, geodesic viewsheds, least-cost paths, DEM and slope analysis, raster reclassification, weighted cost surfaces, map layouts

Programming: Python, Java, SQL, JavaScript, HTML, CSS; object-oriented programming, tree data structures, recursion, algorithm design

Data Analysis: pandas, NumPy, data cleaning and wrangling, descriptive statistics, regression, Excel (Pivot Tables, VLOOKUP)

Visualization: Matplotlib, Seaborn, Excel charts, interactive treemaps, spatial data visualization

Application & Web Development: React, JavaFX, Pygame, Ollama API, WordPress, Elementor, responsive design

Certifications

IBM Data Analyst Professional Certificate | *IBM (Coursera)* In Progress

CS50P – Introduction to Programming with Python | *Harvard University (edX)* Apr 2024

Schulich Business Excellence Academy (SBEA) | *Schulich School of Business* Jul 2022